

Question 1

```
Enter the number of elements: 10
Enter 10 elements (first element is the bottom):
12
23
56
44
99
44
33
55
44
22

Maximum value: 99
Minimum value: 12
Middle elements (even count): 99 and 44
```

Question 2

```
Enter size of Stack 1: 3
Enter elements for Stack 1 (bottom-to-top order):
22
26
27
Enter size of Stack 2: 4
Enter elements for Stack 2 (bottom-to-top order):
77
44
99
44

Circular Queue elements: 22 26 27 44 44 77 99
```

Question 3

```
--- Double-Angle Formula RHS ---
Infix:  ± sqrt ( ( 1 - cos(θ) ) / ( 1 + cos(θ) ) )
Postfix: 1 cos(θ) - 1 cos(θ) + / sqrt ±
Prefix:  ± sqrt / - 1 cos(θ) + 1 cos(θ)

--- Pythagorean Identity RHS ---
Infix:  1 - cos(θ) ^ 2
Postfix: 1 cos(θ) 2 ^ -
Prefix:  - 1 ^ cos(θ) 2

--- Product-to-Sum Identity RHS ---
Infix:  ( 1 / 2 ) * [ cos(α-β) - cos(α+β) ]
Postfix: 1 2 / cos(α-β) cos(α+β) - *
Prefix:  * / 1 2 - cos(α-β) cos(α+β)
```

Question 4

```
Enter Expression 1: x^2 + 2*x + 1
Enter Condition 1: x >= 0
Enter Expression 2: 2*x^2 - x + 3
Enter Condition 2: x < 0
Enter x: 5

Output:
The program selects:
Condition: x >= 0
Expression: x^2 + 2*x + 1
Postfix Expression: x 2 ^ 2 x * + 1 +
Calculated Value: 36
```

Question 5

Enter the number of terms n: 3

Series: 1 + 4 + 7

Postfix: 1 4 + 7 +

Prefix: + + 1 4 7